☕ Coffee Quality & Production Analysis Using SQL

Project Overview

This project analyzes a global coffee quality dataset using SQL. The objective is to identify the best coffee-producing countries, top-performing coffee varieties, processing methods, and production trends. The analysis helps businesses make data-driven decisions regarding sourcing, quality control, and market expansion.

Business Problem

Coffee companies need to understand which countries and coffee varieties consistently produce high-quality coffee. This project provides insights into quality scores, production volume, and processing techniques to support strategic business decisions.

Dataset Information

- Total Records: 100
- Countries: Brazil, Ethiopia, Colombia, India, Kenya
- Attributes:
- Country
- Variety
- Processing Method
- Bags Produced
- Total Score

Database Schema

Table: coffee

Column	Data Type
id	INT
country	VARCHAR(50)
variety	VARCHAR(50)
processing_method	VARCHAR(50)
bags	INT

Column	Data Type
total_score	DECIMAL(5,2)

SQL Concepts Used

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- Aggregate Functions
- CASE Statements
- Subqueries
- Window Functions
- RANK()
- DENSE_RANK()

Business Questions Solved

1. Which country produces the highest quality coffee?
2. Which coffee variety performs best?
3. Which processing method gives the highest score?
4. Which countries have the highest production volume?
5. Which coffee samples score above average?
6. What is the relationship between production and quality?

Key Insights

- Kenya achieved the highest average quality score.
- Ethiopia consistently produced premium-quality coffee.
- Washed processing method generated the best quality ratings.
- Brazil led in production volume.
- Bourbon and Heirloom varieties performed exceptionally well.

Tools Used

- MySQL
- SQL
- GitHub

Project Structure

Coffee-Quality-Analysis-SQL-Project

```
├── Dataset
│   └── coffee.csv

├── SQL Files
│   ├── schema.sql
│   ├── insert_data.sql
│   └── analysis_queries.sql

├── Outputs
│   ├── top_countries.png
│   ├── best_varieties.png
│   ├── production_analysis.png
│   └── quality_ranking.png

├── README.md
└── Project_Report.pdf
```

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